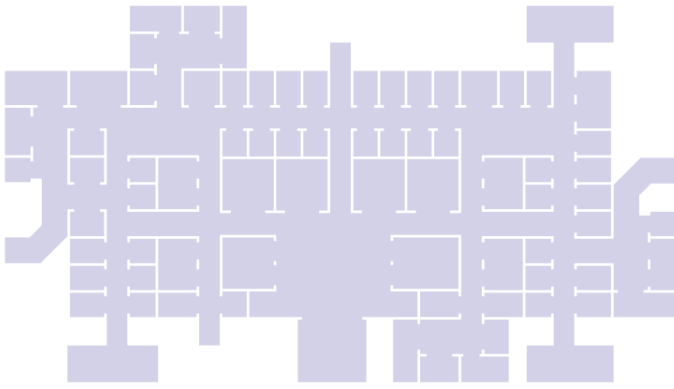
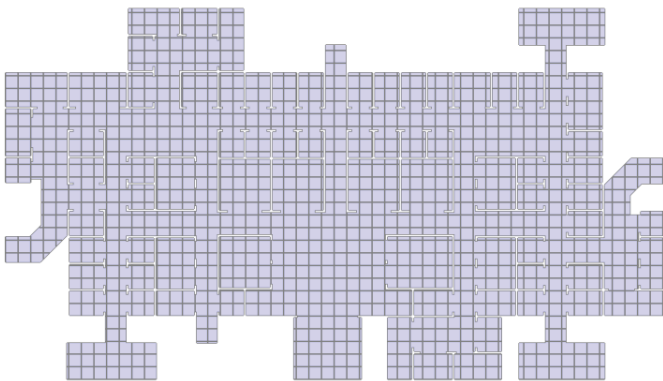




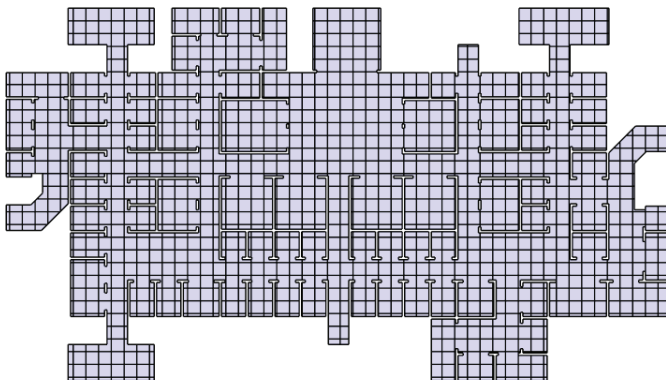
Base Floor Plan

The floor plan of the shopping centre imported as a DXF file, reconstructed as a topological face using perimeter and inner profile boundaries. It serves as the base geometry for all subsequent spatial analysis operations.



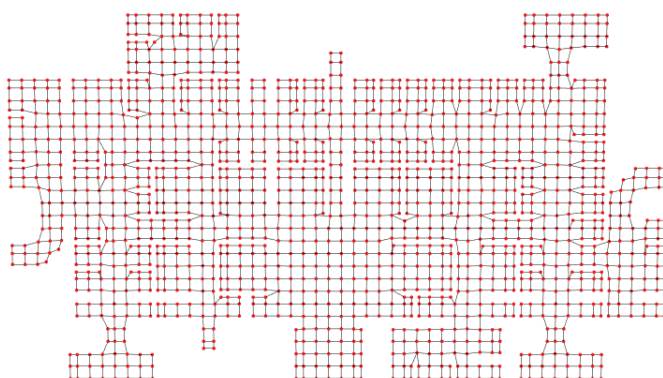
Floor Plan with Grid

A uniform 1.5mx1.5m unit grid overlay applied across the shopping centre floor plan. This grid is used to slice the floor plan into discrete spatial cells for graph-based analysis.



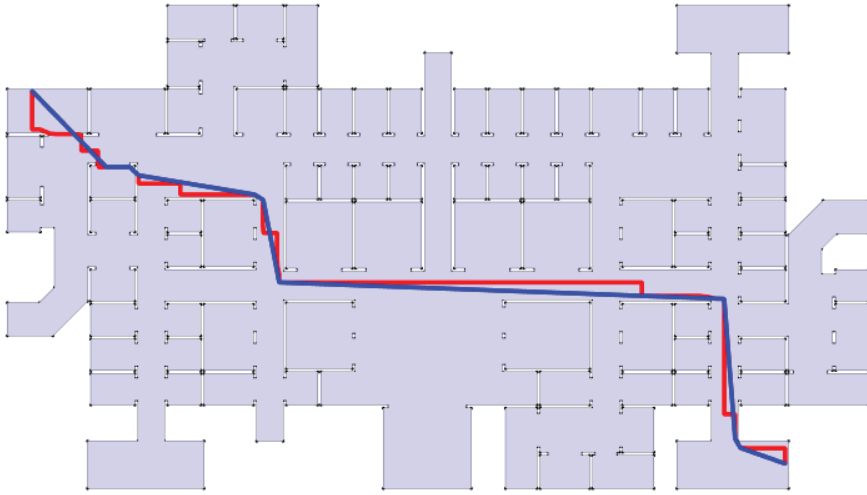
Topological Shell

The floor plan sliced by the grid into a topological shell of individual face cells, each assigned a unique face_id for dictionary tracking. This discretized representation forms the basis for deriving the analysis graph.



Analysis Graph

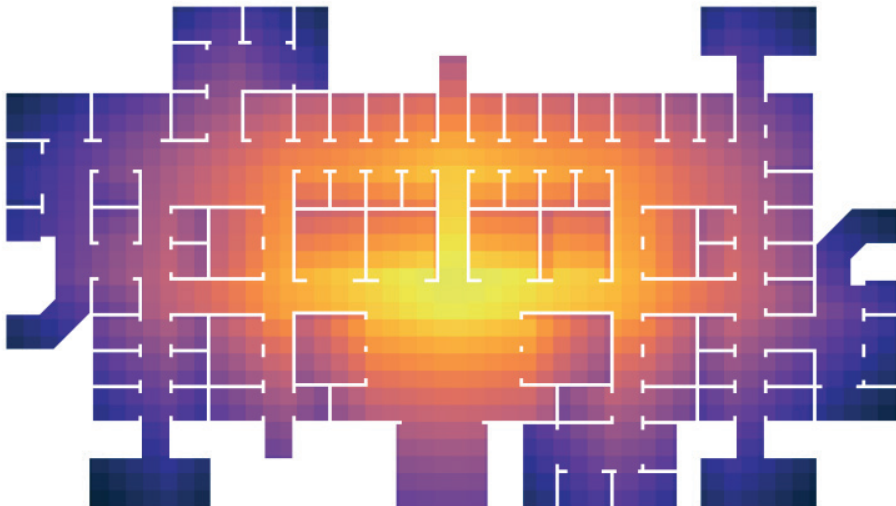
The adjacency graph where each node represents a grid cell and edges represent shared boundaries between adjacent cells. This graph encodes the spatial connectivity structure of the entire shopping centre.



Shortest Path

The shortest path computed between the upper-left and lower-right corners of the floor plan on a compiled routing graph.

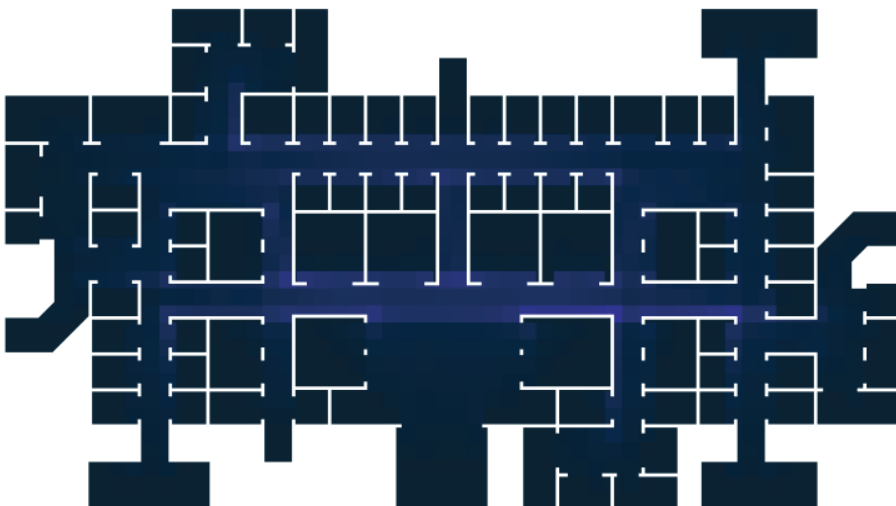
The red path shows the raw shortest route while the blue path shows the straightened optimised route. The longest blue segment highlights a spine with a long reach and identifying such core routes would help to distribute crowds, manage easy circulation and attract primary shopping zones



Closeness Centrality

Closeness centrality measured how quickly each spatial cell can reach all other cells in the floor plan.

The bright central zone confirms that the main corridors running through the heart of the shopping centre is the most spatially integrated area.



Betweenness Centrality

Betweenness centrality identified cells that act as critical passage points on shortest routes between all pairs of spaces.

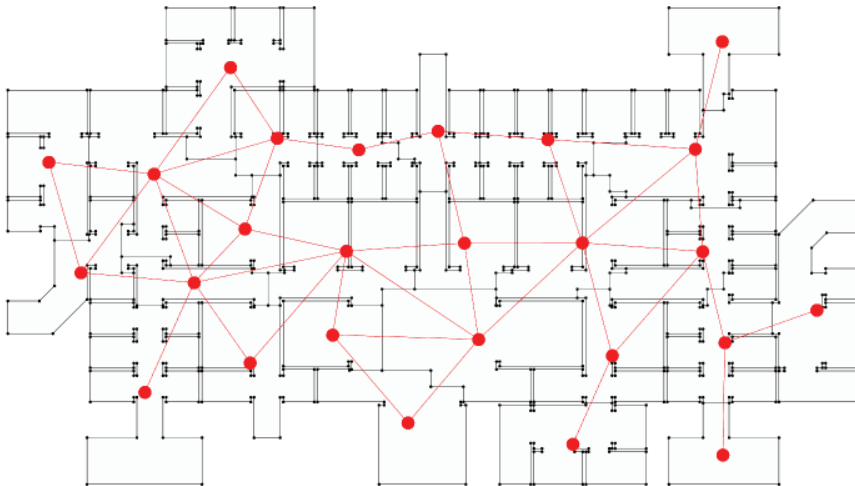
The highlighted corridor spine indicates it is the primary bottleneck and core movement channel of the shopping centre.



Community Detection

Spatial communities identified within the shopping centre reveal distinct spatial zones through color coding.

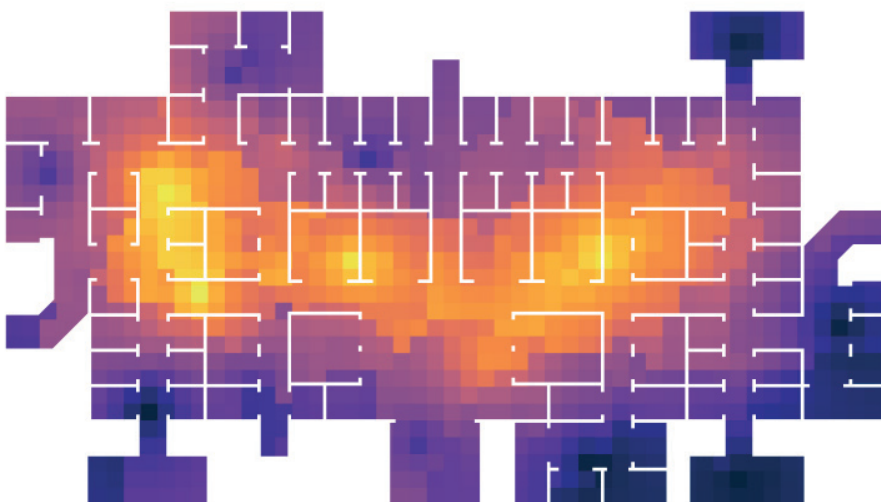
The color coded diagram clearly shows a highlight shopping zones, easy accessible zones along the spines and clustered specialized zones distributed in a branched structure



Community Graph

A simplified graph built from the external boundaries of each detected community, showing inter-community connectivity.

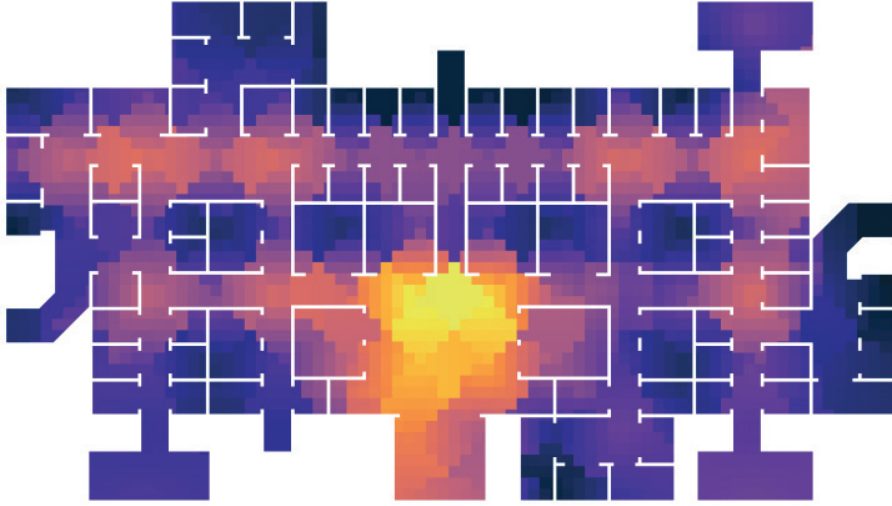
This abstracted representation highlights how the major spatial zones of the shopping centre relate and connect to each other and identifies specialized isolated branches



Degree Centrality

Degree centrality mapped across the floor plan showing how many direct neighbours each cell has, with warmer colours indicating higher connectivity.

The corridor connections and central open areas of the shopping floor show the highest degree, confirming them as the most directly connected spaces in the layout.



Visibility Analysis

Isovist-based visibility analysis computed using `Face.Isovist()`, measuring how many spatial cells are visible from each viewpoint across the floor plan.

The bright hotspot in the central atrium area reveals the single most visually dominant position in the shopping centre

The bigger shops aligned to the visibly dominant areas reflect good design thinking, whereas some smaller shops could be merged to create larger ones to exploit more visibility and traction